Security Patrol Robot

Autonomous patrol, intrusion detection & avoidance on iRobot Create3

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Outline

1. Project Idea
2. What Was Achieved
3. Demo
4. Architecture
5. Difficulties & Observations
6. What Would Be Next
7. The Team

Initial Project Plan

- **Concept:** autonomous security guard robot
- Patrols a square perimeter for N laps
- Detects intrusions via bump sensors → triggers alarm → avoids obstacle → resumes patrol
- After all laps → returns to dock
- Manual override via service calls

Platform: iRobot Create3 with ROS 2

What Was Achieved

Fully working:

- FSM with 8 states
- Undocking / Docking actions
- Square patrol with lap counting
- Bump detection $\rightarrow$ siren $\rightarrow$ avoidance
- Avoidance maneuver
- Manual override via ROS 2 service calls (/patrol/stop, /patrol/start)

Partial / not completed:

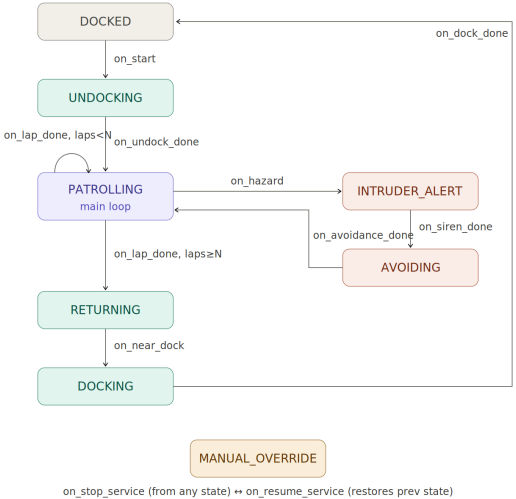
- Gazebo simulation testing

Demo

Live demo

1. Start patrol: `ros2 service call /patrol/start std_srvs/srv/Trigger`
2. Robot undocks → patrols square → bump triggers siren → avoids → resumes
3. Stop: `ros2 service call /patrol/stop std_srvs/srv/Trigger`

FSM State Machine



Code Organization :`patrol_fsm`

`patrol_node.py`	Main ROS 2 node: action clients (undock, dock, draw_square, audio), hazard/odom subscribers, start/stop services, state dispatcher
`fsm_states.py`	Pure Python FSM + transition methods -> zero ROS dependency, testable
`avoidance.py`	3 phase odom maneuver
`siren.py`	Single function `build_siren_goal()`: 880/659 Hz alternating tones, ∼3 s
`launch/`	Launch files

ROS 2 Interfaces

Actions used (client):

- `/Robot2/undock`
- `/Robot2/dock`
- `/draw_square`
- `/Robot2/audio_note_sequence`

Topics & Services:

- Sub: `/Robot2/hazard_detection`
- Sub: `/Robot2/odom`
- Pub: `/Robot2/cmd_vel`
- Srv: `/patrol/start`, `/patrol/stop`

Avoidance Maneuver : 3 Phases

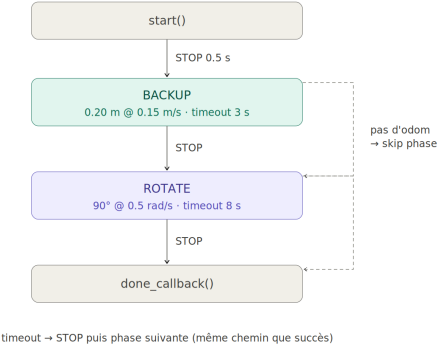


Figure: Robot avoidance maneuver

Difficulties & Observations

- **Create3 bump reflex**: robot has a reflexe that fight `cmd_vel` right after a bump -> need to wait before sending backup command
- **QoS mismatch**: Create3 sensors use `BEST_EFFORT` and default `RELIABLE` silently drops all messages
- **Concurrency**: needed `ReentrantCallbackGroup` + `MultiThreadedExecutor` avoid single threaded executor deadlocks on nested action calls
- **Turtlesim $\neq$ real robot**: odometry, timing, and physics differ significantly
- **Lab issues**: network issues, debugging on real hardware is slow

What Would Be Next

- **LIDAR detection**: replace bump sensor with LIDAR for earlier detection
- **Configurable patrol path**: arbitrary waypoints instead of fixed square
- **Camera integration**: intruder identification / logging
- **Web dashboard**: real time state monitoring, intruder notifications, remote start/stop

The Team

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Thank you! Questions?